

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

03rd May 2016

Subject SA1 – Health and Care Insurance

Time allowed: Three hours (14.45* - 18.00 Hrs)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
4. *The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which are meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** Softech systems India is a large technology company with more than 3 Lakhs employees. The company provides health and wellness benefits to its employees by self-funding. The fund is used to provide unlimited in-patient, out-patient hospital benefits and preventive care/wellness benefits to the employees and their immediate family dependents (excluding Parents).

In the last few years, the contribution levels have been increasing and have become very volatile. The Finance Director is worried about the levels of uncertainty and has approached your firm with an assignment.

You are a newly qualified health actuary employed with the firm and your manager has asked you to work on this assignment.

- i)** Why would a company decide to self-insure the benefits? (4)

- ii)** Briefly explain the probable reasons for the volatility in the contributions levels and possible mitigation techniques available to the Finance Director (6)

The finance director wants to insure some of the benefits and feels that the company can control the volatility by taking insurance.

- iii)** Discuss which form of insurance will be appropriate and how it can be structured? (8)

The Finance Director intends to get quotations from major health insurers for a stop loss cover

- iv)** Assuming you are the actuary working with a health insurer, list the information you will need from the ceding company and discuss the aspect that you will need to consider before deciding whether to quote or not. (7)
- [25]**

- Q. 2)** Healing Health Insurance is a large health insurer and has been selling health insurance products for the last 8 years. The company largely sells health indemnity and standalone Critical Illness plans. The premium rates are yearly reviewable.

The company is now exploring introducing a severity based Critical Illness product that would pay depending the severity of the covered critical illness.

- i)** Why should the company launch this product? (5)

- ii)** List the main risks of launching a new product? (5)

- iii)** Discuss the mains considerations in the areas of

a) Product design & pricing,

b) Underwriting and Claims,

c) Premium reviewability

d) Sales and other considerations

(20)

The marketing department has proposed to introduce multi-pay CI whereby similar illnesses will be clubbed in different buckets. For example, all cardiovascular diseases like heart attack, heart organ transplant, CABG will be put under one bucket and other related conditions will be put in other buckets.

There are 3 such buckets proposed by the team. The benefit will be payable only once from each of the buckets. The Sum Assured will also be different for each bucket. The premiums will be level for the term and all future premiums will be waived off once the first claim is made. The policy however will continue till all 3 claims are paid or 10 years whichever is earlier. The premiums will be guaranteed for the first 5 policy years and then reviewable at 5 year intervals. The plan however will be guaranteed renewable for the lifetime of the policyholder or until all 3 CI claims (one from each bucket) are paid.

The marketing team has further proposed that the applicant should have the option to choose the buckets at the time of application. For example, the applicant should be able to choose either one bucket or all three buckets at the time of application and pay premiums accordingly.

- iv)** Briefly evaluate the various aspects of the proposal. (10)
[40]

Q. 3) You are a nearly qualified actuary working with an established non-life insurance company selling only non-health products. The company is selling non-life insurance policies for last ten years and has major exposure on motor and fire lines of business. As a part of its business expansion strategy, it wants to foray into health insurance space.

- i)** What would be advantages and dis-advantages for the company to move into health insurance space? (8)
- ii)** Given the company has no relevant past experience with it, how it will deal with listed requirements for health insurance business:
- a)** Asset Liability Management
 - b)** Capital and Solvency assessment
 - c)** Claims and expense management (15)

[23]

Q. 4) A health insurance company, in line with the current market practice, uses only age and gender as the rating factors for pricing their individual health indemnity product.

In order to increase the sales, the company wants to further differentiate the premium by the smoking status of the individual so that it can offer cheaper premiums to non-smokers. Since the other companies don't currently have smoker differentiated rates, the management is confident that this will help in increasing the sales. The smoking status is not currently captured by the company.

You are a qualified actuary and have been asked by your appointed actuary to provide your recommendation on this proposal. Discuss the main aspects of the proposals that need to be considered. [12]
